

DB schema

Data will not be save in multiple tables as Multi-table DynamoDB is not the best practice. Instead, Data will be logically divided into 10 groups. When exporting from DynamoDB, the data will be exported in JSON format and saved in that logical grouping. (Graphs below & sample record)

Why JSON ?

1. Logical grouping

Data is organized into clearly defined groups, making it easier to understand and manage. Each section in the JSON corresponds to a logical part of the transaction, which keeps the structure clean and intuitive.

2. Standardized format

JSON is a universally accepted data format, supported by almost every programming language and system. This ensures smooth integration across different tools and platforms.

3. Easier for frontend development

The frontend team can directly use the grouped data without extra processing. Each group in the JSON can map directly to frontend components, speeding up development.

4. Easier for administrators to share data

Admins can easily export and share the JSON file without needing complex tools. JSON is lightweight and readable, making it ideal for audits, reporting, and external communication.

5. Extensible and flexible

As requirements grow, new fields or groups can be added to the JSON without changing the database design. This keeps your system adaptable and future-proof.

Table name:

TransactionsTable

Partition Key (PK):

PORT#<CustomsOfficeCode>

→ This groups all transactions by Customs Office. Useful when filtering data for a specific office.

Sort Key (SK):

DATE#<RegistrationDate>#TX#<ReferenceNumber>

→ Helps to query by date and unique reference number inside that customs office. More GSI can be added

Attributes Grouping

G1 (identification) INDEX Year Regime Registration Serial Registration Number Reference Number Registration Date Registration Time		G2 (declarant) Declarant CR Declarant Name Agent ID Agent Name Consignee CR Consignee Name Exporter CR Exporter Name
G4 (port info / location) Customs Office Code Customs Office Name LOC Code LOC Name Office Exit Office Exit Name Exit Office Code	G3 (payments) Receipt Serial Receipt Number Receipt Date Cashier ID Cashier Name	G5 (cargo) Item Number Procedure CP3 Pref HSCode Commercial Description Specification Code Warehouse Code

G6 (country): Country of Origin Code Country of Origin Country of Export Code Country of Export Country of Destination Country of Destination	G7 (financials): Invoice Currency Invoice Amount Local Currency Local Amount Customs Duty Rate Customs Duty BHD Excise Duty Rate VAT Rate BHD Total Duty BHD HS-Rate
G8 (logistics): Gross Weight Net Weight Package Code Package Amt Sup Unit Sup Amt	G9 (exit details): Exit Serial Exit Number Exit Date Exit Time Exit Officer ID
	G10 (status): Status Assigned Examiner First Reroute By

- G1 (identification):

INDEX
Year
Regime
Registration Serial
Registration Number
Reference Number
Registration Date
Registration Time

- G2 (declarant):

Declarant CR
Declarant Name
Agent ID
Agent Name
Consignee CR
Consignee Name
Exporter CR
Exporter Name

- G3 (payments):

Receipt Serial
Receipt Number
Receipt Date
Receipt Time
Cashier ID
Cashier Name

- G4 (port info / Location):

Customs Office Code
Customs Office Name
LOC Code
LOC Name
Office Exit
Office Exit Name
Exit Office Code

- G5 (cargo):

Item Number
Procedure
CP3
Pref
HSCode
Commercial Description
Specification Code
Warehouse Code

- G6 (country):

Country of Origin Code
Country of Origin
Country of Export Code
Country of Export
Country of Destination Code
Country of Destination

- G7 (financials):

Invoice Currency
Invoice Amount
Local Currency
Local Amount
Customs Duty Rate
Customs Duty BHD
Excise Duty Rate
Excise Duty BHD
VAT Rate
VAT BHD
Total Duty BHD
HS-Rate
Fees
Currency

- G8 (logistics):

Gross Weight
Net Weight
Package Code
Package Amt
Sup Unit
Sup Amt

- G9 (exit details):

Exit Serial
Exit Number
Exit Date
Exit Time
Exit Officer ID
Exit Officer Name

- G10 (status):

Status
Assigned Examiner
First Reroute By

JSON Sample record

```
{
  "PK": "PORT#2101",
  "SK": "DATE#20190102#TX#448",
  "identification": {
    "index": "2019-01-EX-1",
    "year": 2018,
    "regime": "EX",
    "registrationSerial": 1,
    "registrationNumber": 15,
    "referenceNumber": 448,
    "registrationDate": "2019-01-02",
    "registrationTime": "08:14:19"
  },
  "parties": {
    "declarantCR": "123456789",
    "declarantName": "ABC Importers",
    "agentID": "AG789",
    "agentName": "Global Agents Ltd.",
    "consigneeCR": "456789123",
    "consigneeName": "XYZ Consignee Co.",
    "exporterCR": "789456123",
    "exporterName": "Exporter Global Inc."
  },
  "payment": {
    "receiptSerial": "RS-2019",
    "receiptNumber": "RN-448",
    "receiptDate": "2019-01-02",
    "receiptTime": "09:00:00",
    "cashierID": "CASH123",
    "cashierName": "John Doe"
  },
  "location": {
    "customsOfficeCode": 2101,
```

```
"customsOfficeName": "Khalifa Bin Salman Port",
"locCode": "LOC001",
"locName": "Main Warehouse",
"officeExit": "EX001",
"officeExitName": "Main Exit Gate",
"exitOfficeCode": "2101"
},
"itemCargo": {
  "itemNumber": 1,
  "procedure": "Export",
  "cp3": "CP3-XYZ",
  "pref": "PREF-ABC",
  "hsCode": "123456",
  "commercialDescription": "Electronics - Mobile Phones",
  "specificationCode": "SPEC123",
  "warehouseCode": "WH001"
},
"countryDetails": {
  "countryOfOriginCode": "CN",
  "countryOfOrigin": "China",
  "countryOfExportCode": "CN",
  "countryOfExport": "China",
  "countryOfDestinationCode": "BH",
  "countryOfDestination": "Bahrain"
},
"financials": {
  "invoiceCurrency": "USD",
  "invoiceAmount": 10000,
  "localCurrency": "BHD",
  "localAmount": 3775,
  "customsDutyRate": 5,
  "customsDutyBHD": 150,
  "exciseDutyRate": 0,
  "exciseDutyBHD": 0,
```

```
"vatRate": 5,
"vatBHD": 200,
"totalDutyBHD": 350,
"hsRate": 5,
"fees": 50,
"currency": "BHD"
},
"logistics": {
  "grossWeight": 500,
  "netWeight": 480,
  "packageCode": "PKG001",
  "packageAmt": 10,
  "supUnit": "Boxes",
  "supAmt": 10
},
"exitDetails": {
  "exitSerial": "EX-SERIAL-01",
  "exitNumber": "EX-NUM-448",
  "exitDate": "2019-01-02",
  "exitTime": "09:32:00",
  "exitOfficerID": "771002696",
  "exitOfficerName": "Inspector Ali"
},
"statusInfo": {
  "status": "Released",
  "assignedExaminer": "Salman Hasan",
  "firstRerouteBy": "Hani Ramadan"
}
}
```